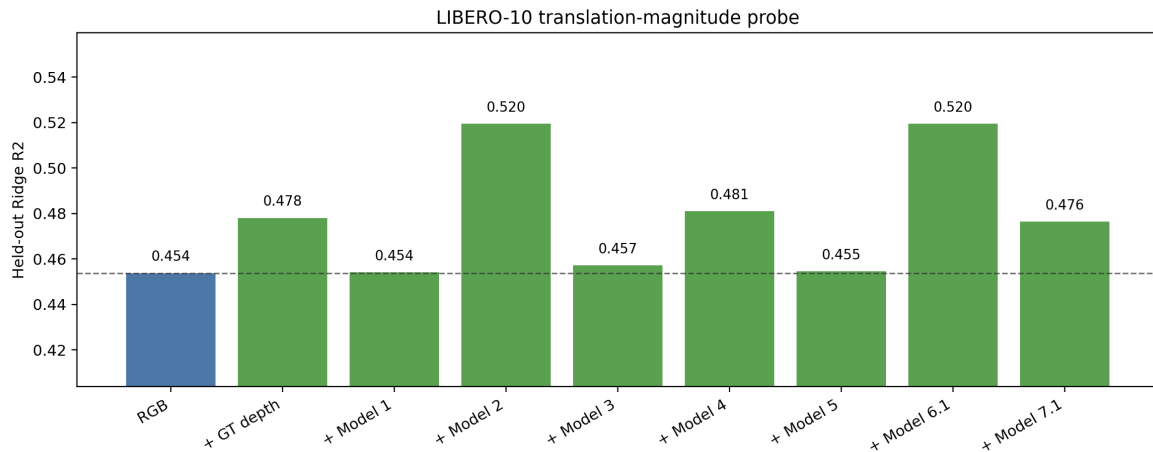


Redone Feature Analysis for D-LAPA Depth Injection

Inputs: local paper *Depth Information Injection in VLA with Latent Action Pretraining via Synthetic Depth Data*, the prior feature-analysis appendix, and the provided `libero10_features` cache.

Protocol: 138,090 LIBERO-10 frame pairs, video-disjoint 80/20 split with seed 42 (111,284 train, 26,806 held out). Features are z-scored using train statistics only. The probe is Ridge regression with $\alpha=1000$ and the LSQR solver, evaluated on xyz translation magnitude.

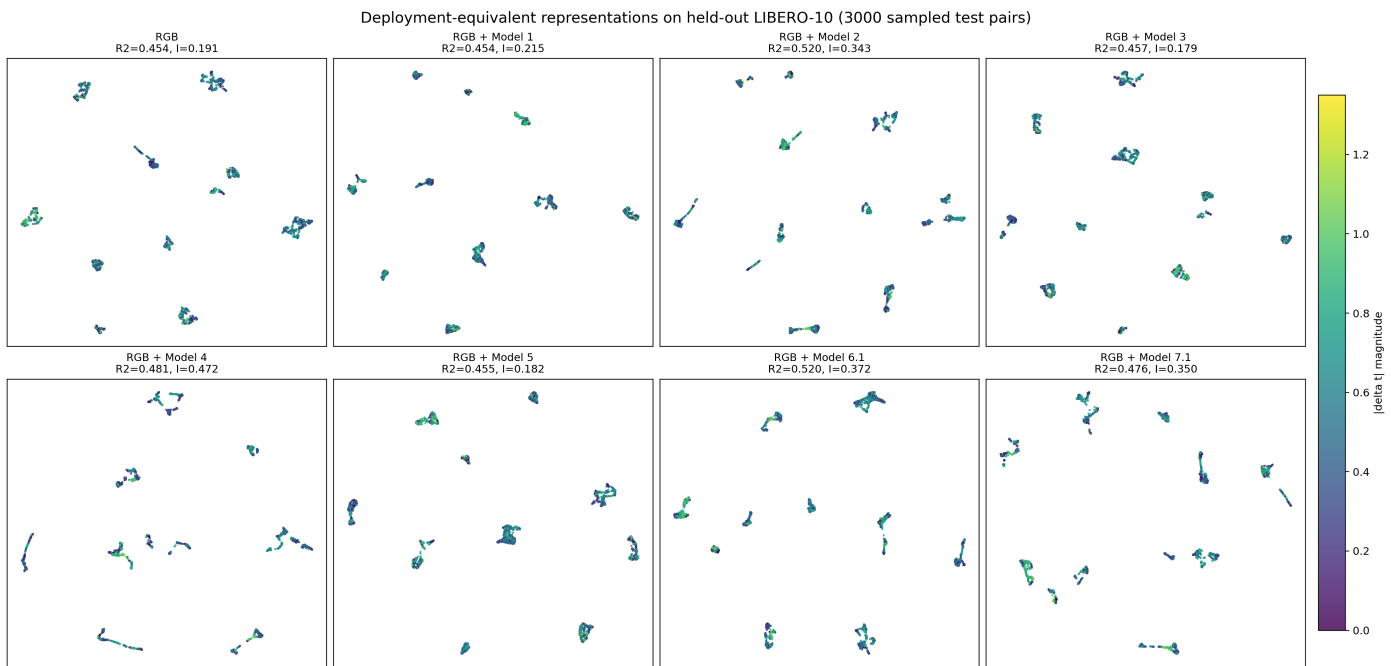
Cache compatibility note. The provided `z_rgb_feature_input` behaves like the paper's pre-VQ LAPA-LAQ reference ($R^2=0.454$), not the earlier appendix's finetuned-LAPA RGB reference ($R^2=0.619$). The old figure should therefore be relabeled or rerun with a finetuned-LAPA feature cache if that baseline is required.



Representation	Dim	R2	Delta	Spearman
RGB	4096	0.4539	+0.0000	0.6826
RGB + GT depth	5120	0.4781	+0.0243	0.7005
RGB + Model 1	5120	0.4541	+0.0002	0.6835
RGB + Model 2	5120	0.5195	+0.0657	0.7264
RGB + Model 3	5120	0.4572	+0.0033	0.6844
RGB + Model 4	5120	0.4811	+0.0273	0.7002
RGB + Model 5	5120	0.4545	+0.0007	0.6832
RGB + Model 6.1	5120	0.5195	+0.0656	0.7249
RGB + Model 7.1	5120	0.4764	+0.0225	0.6962

UMAP Smoothness Snapshot

UMAP is used only as a qualitative visualization on 3,000 sampled held-out pairs. Moran's I measures whether nearby projected points share similar action magnitude. The quantitative conclusions should be read from the probe table.



Representation	Moran's I
RGB	0.1906
RGB + Model 1	0.2154
RGB + Model 2	0.3435
RGB + Model 3	0.1793
RGB + Model 4	0.4716
RGB + Model 5	0.1815
RGB + Model 6.1	0.3723
RGB + Model 7.1	0.3497

Interpretation: Model 2 and Model 6.1 give the strongest linear magnitude decodability ($R^2=0.520$), while Model 4 gives the smoothest UMAP magnitude neighborhoods (Moran's $I=0.472$). These are complementary diagnostics, not identical claims.

Diagnostics and Replacement Text

Depth-feature faithfulness to the Stage-1 depth teacher:

Feature	MSE to GT	Cosine to GT
Model 1	0.90801	-0.0073
Model 2	0.89051	0.0228
Model 3	1.03059	0.0104
Model 4	0.14223	0.0066
Model 5	0.13026	-0.0030
Model 6.1	0.90316	-0.0030
Model 7.1	0.02497	0.8060
GT depth	0.00000	1.0000

Depth-index diagnostics where predicted indices/confidence are available:

Feature	Token acc.	Seq. acc.	Mean conf.
Model 1	0.1234	0.0010	0.3872
Model 2	0.1058	0.0003	0.4012
Model 3	0.1295	0.0002	0.6798
Model 6.1	0.2818	0.0041	0.3848

Suggested replacement feature-analysis paragraph:

We probe the geometric content of deployment-equivalent representations on 138,090 LIBERO-10 frame pairs using a video-disjoint 80/20 split. The provided RGB representation reaches $R^2=0.454$, indicating that this cache corresponds to the pre-VQ LAPA-LAQ feature scale rather than the finetuned-LAPA reference used in the earlier appendix. Concatenating Stage-2.5 depth features gives a selective improvement: RGB + Model 2 reaches $R^2=0.520$, a +0.066 absolute gain over RGB; Model 6.1 is effectively tied at $R^2=0.520$. The no-depth controls Model 3 and Model 5 remain close to RGB (+0.003 and +0.001), supporting the interpretation that the main gain comes from explicit depth-image variants rather than capacity alone. Model 4 is weaker in linear R^2 but produces the smoothest UMAP magnitude neighborhoods, so the feature analysis should separate linear decodability, visual smoothness, and downstream policy success.

Main conclusion: depth-derived representations add measurable geometric decodability to the provided pre-finetune RGB feature cache, but the result is incremental and must be reported with the cache mismatch and downstream policy evaluation caveats.